1 Problem Validation

Before starting the design, we conducted a problem validation exercise. This was done using a structured questionnaire distributed in the municipal council area.

1.1 Questionnaire Design

The questionnaire contained 13 questions covering the following areas:

- The type of streetlights currently in use.
- How streetlights are currently switched on and off.
- How often lights remain on during the day.
- How often faults occur and what types of faults are common.
- How faults are currently detected and how long detection takes.
- The problems caused by delayed fault detection.
- Whether stakeholders believe an automated system would be useful.
- Their concerns about such a system.

1.2 Key Findings

The responses to the questionnaire confirmed the following:

- A significant number of respondents reported that streetlights staying on during the daytime was either a frequent or occasional occurrence in their area.
- Streetlight faults were reported to occur often, with bulb failure is being the most common fault types.
- Most faults were detected through public complaints and routine inspections, rather than through any automated system.
- In many cases, identifying a faulty streetlight took more than one day.
- The main concerns raised by respondents were installation cost, reliability, and maintenance complexity.

1.3 Conclusion from Validation

The validation survey confirmed that the problems we identified are real and significant. It also confirmed that the proposed solution would be welcomed by the people who manage and depend on streetlights. The concerns raised about cost and reliability guided our design decisions, particularly our choice to use low-cost, widely available components and a simple but robust circuit design.